(a) From single-time language-guided segmentation

Open-vocabulary Segmentation

building



Referring segmentation

the white car near the tree



Reasoning Segmentation

the object used for sitting



All rely on a single image.

(b) What is Multi-temporal Referring Segmentation?

Image Input :





Language Query:

Segment the parked white car next to the sidewalk that disappears.

Mask Output:



cross-temporal comparison

(c) Why do we need Multi-temporal Referring Segmentation?

Urban Monitoring

the red vehicle along the roundabout that disappeared canvas



Environmental Change

the forested region that disappeared



Street Objects Change

the trash can close to the house s corner that disappeared



Real-world changes are temporal